

LLM/NLP 101

Large Language Models (LLMs) – Natural Language Processing (NLP)

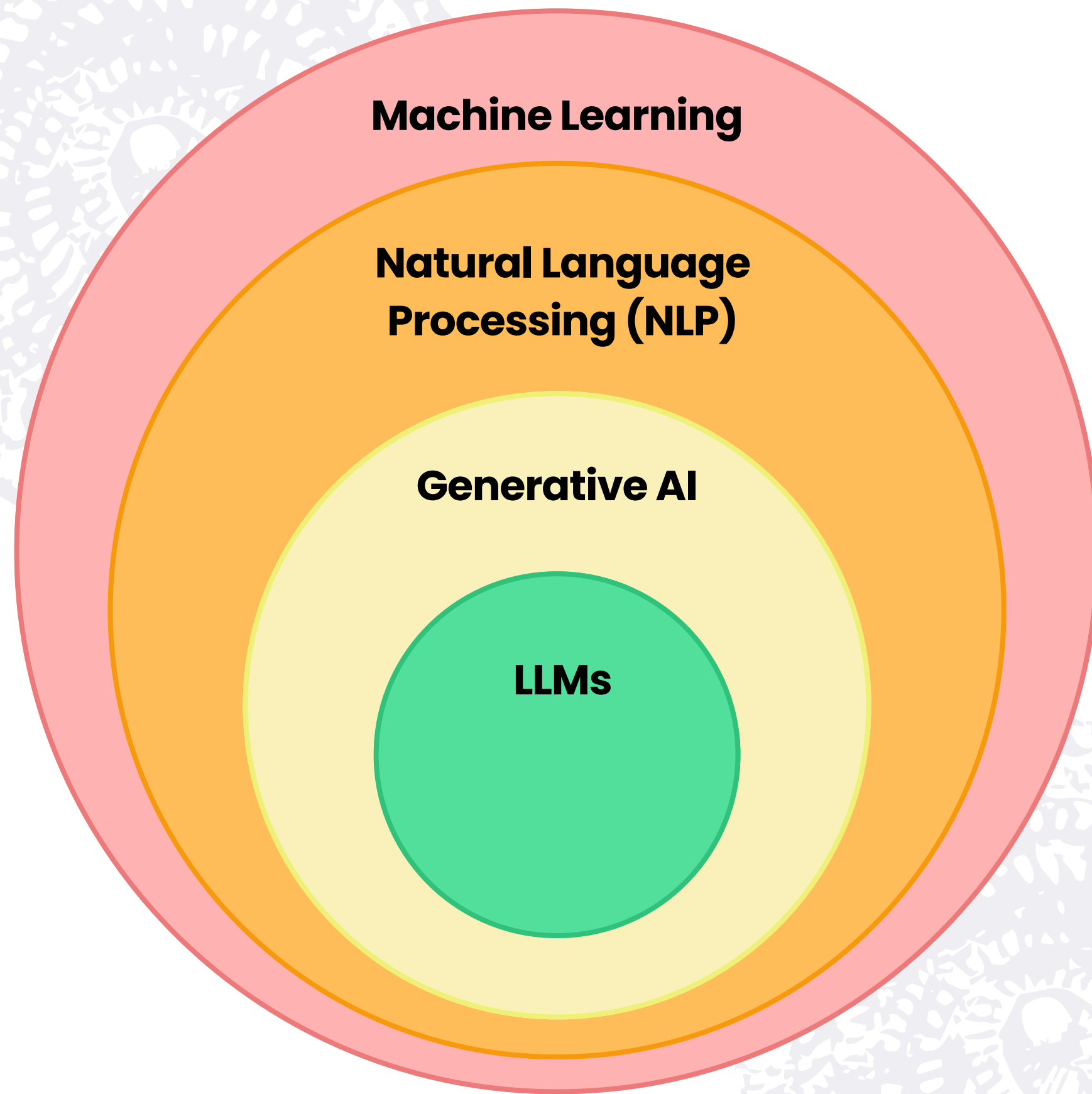
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Overview

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Machine Learning

A way of building AI. Systems learn patterns from data instead of being explicitly programmed rule-by-rule

Natural Language Processing (NLP)

ML applied specifically to language. Focused on getting computers to read, understand, and produce human language.

Generative AI

Creates new content (audio, text, images, video), rather than just classifying or predicting

Large Language Models (LLMs)

Specific technology behind most generative text AI today (e.g. ChatGPT, Claude, Gemini)

LLM's

Where LLM's play a roll during your day



Drafting & editing emails



Summarizing long documents



Quick how-to help



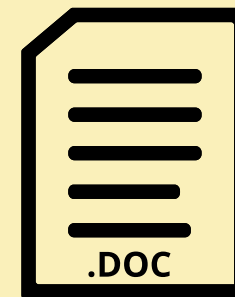
Session prep

Generative AI

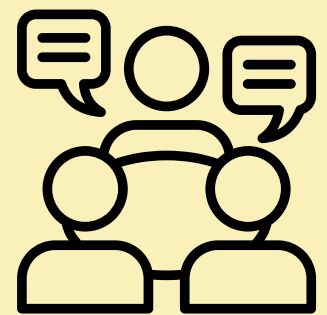
Where Generative AI plays a roll during your day



Social graphics



First-draft copy



Meetings summaries

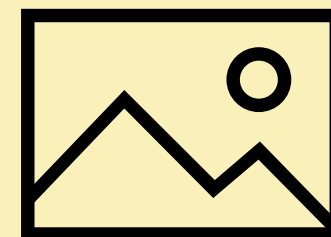
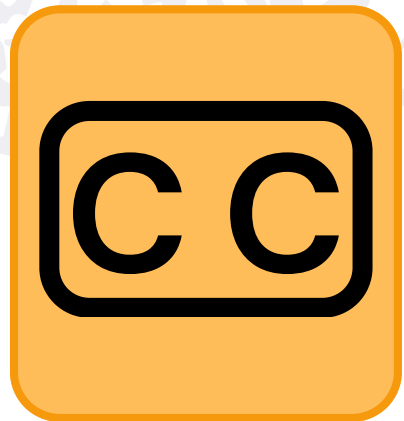


Image generation

NLP

Where NLP plays a roll during your day



Live captions &
transcription



Translation tools



Grammar & spell-check



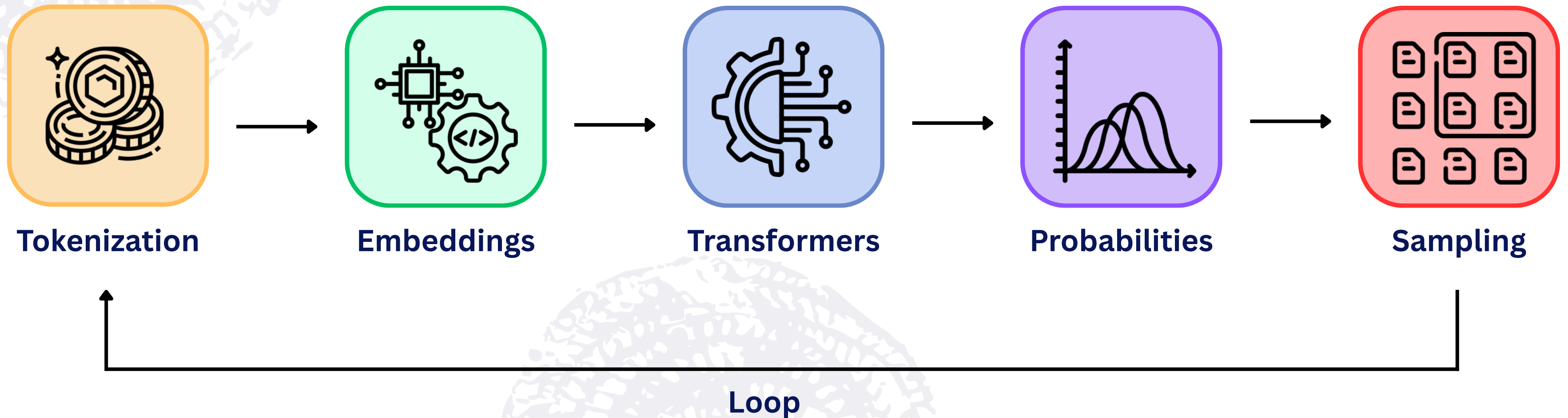
Voice-to-text

How an LLM works

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Overview



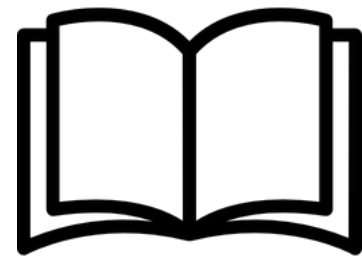
Training Data

PRE-TRAINING

The massive body of data an LLM reads.



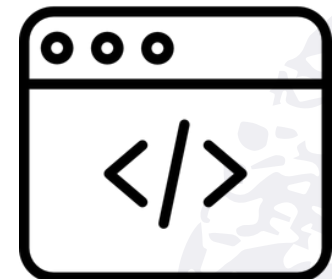
Websites



Books



Articles



Code

It's the model's source of both knowledge and gaps in information.

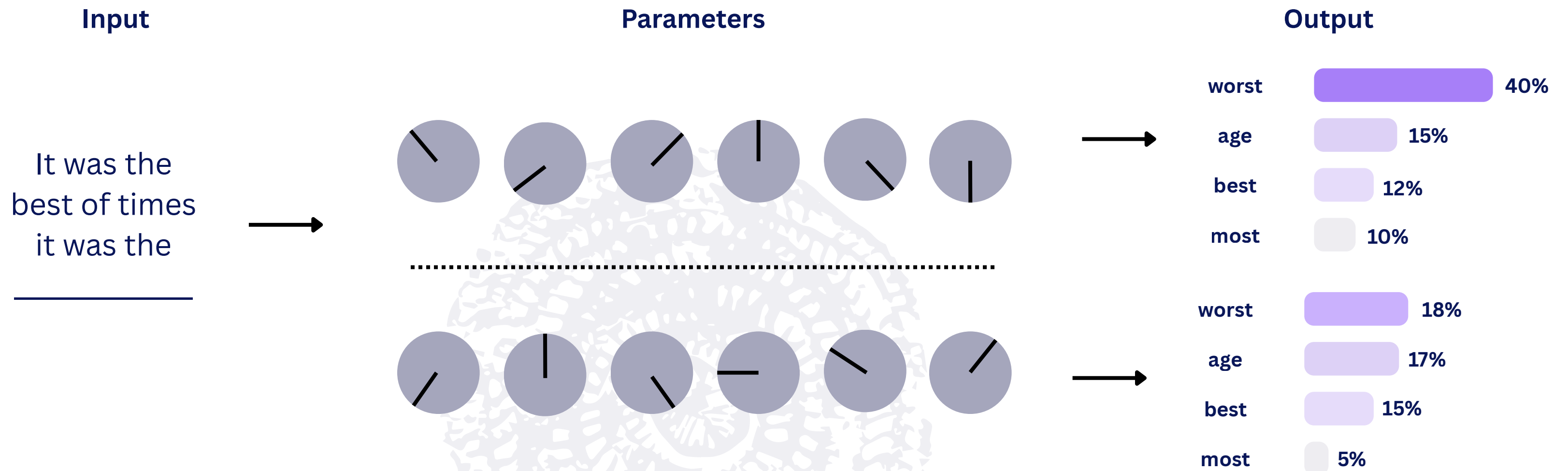
If something is underrepresented in training data (Hawaiian language for example), the model will be weaker on it.

Parameters

PRE-TRAINING

The model's learned settings or internal "knobs" that determine how it weighs and combines information.

Changing these parameters will change the probabilities that the model gives for the next word on a given input.



Pre-Training vs. Inference

Pre-Training

First phase of an AI model. Happens once.

The model reads an enormous amount of text and adjusts its parameters to get better at predicting language.

Inference

Uses the model. Happens in real-time.

Every time you send a message, the model applies what it already learned.

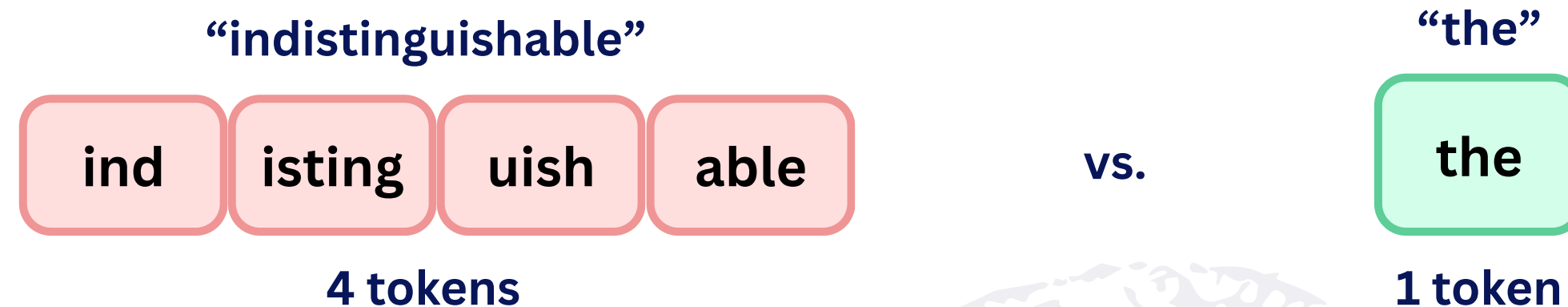
Tokenization

INFERENCE

Before a model can process anything, text is chopped into pieces called **tokens**. Every token is assigned a number (**token ID**).



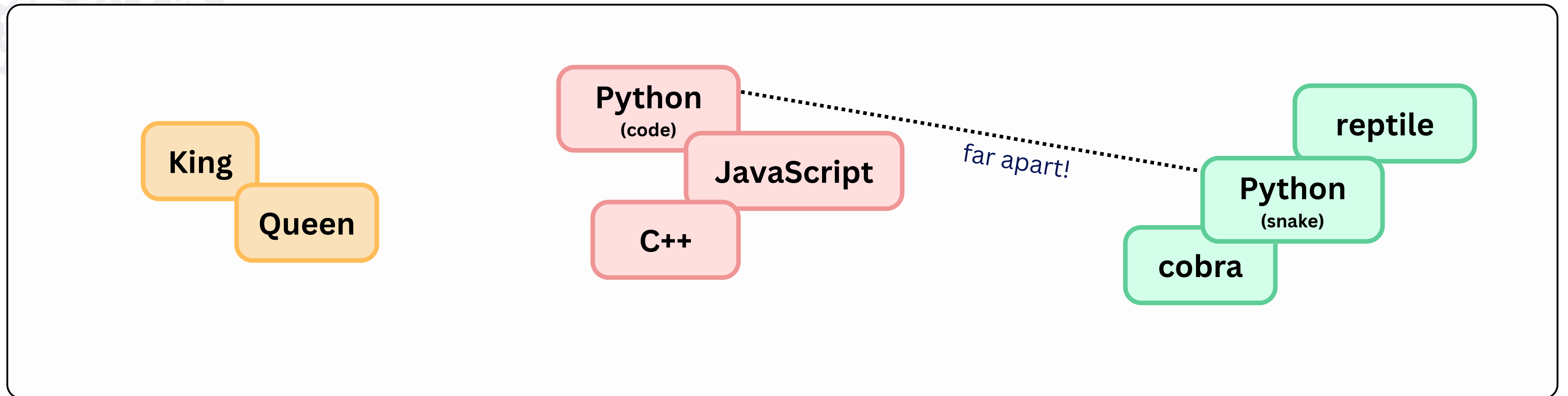
Long or uncommon words get broken into pieces.



Embeddings

INFERENCE

The model turns words into numbers that capture meaning.
Each token becomes a **vector** — coordinates in a "meaning space."



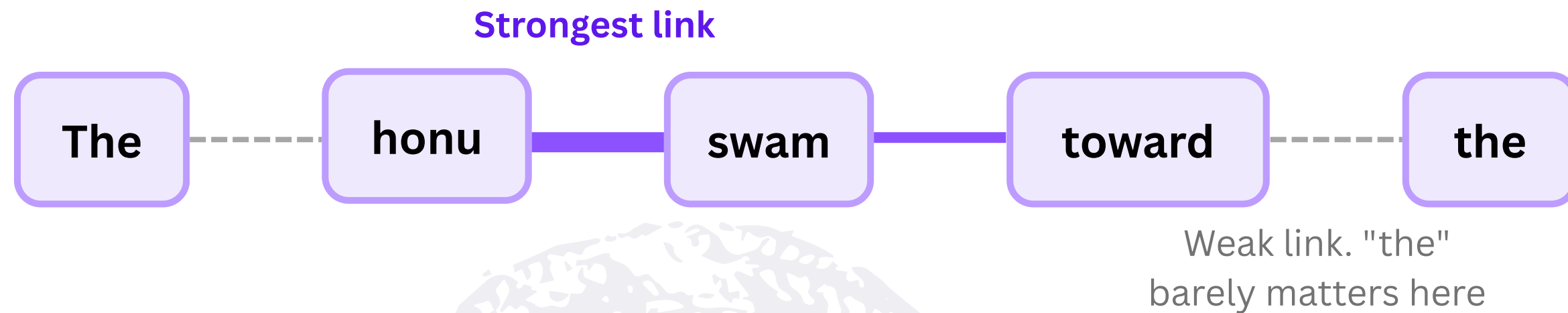
Words used in similar ways end up close together.

That's why "Python" the language sits near "JavaScript," and "Python" the snake sits near "reptile".

Transformers

INFERENCE

For every token, the model decides which tokens matter most.
Thicker lines below mean stronger **attention**.



This happens many times in parallel, through different "**attention heads**." Each one catches a different kind of relationship.

Next-Token Prediction

INFERENCE

Given everything so far, the model guesses the single most likely next token.

The

honu

swam

toward

the



ocean



42%



shore



21%

reef



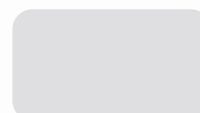
15%

current



12%

sky



10%

Probabilistic vs. Deterministic Sampling

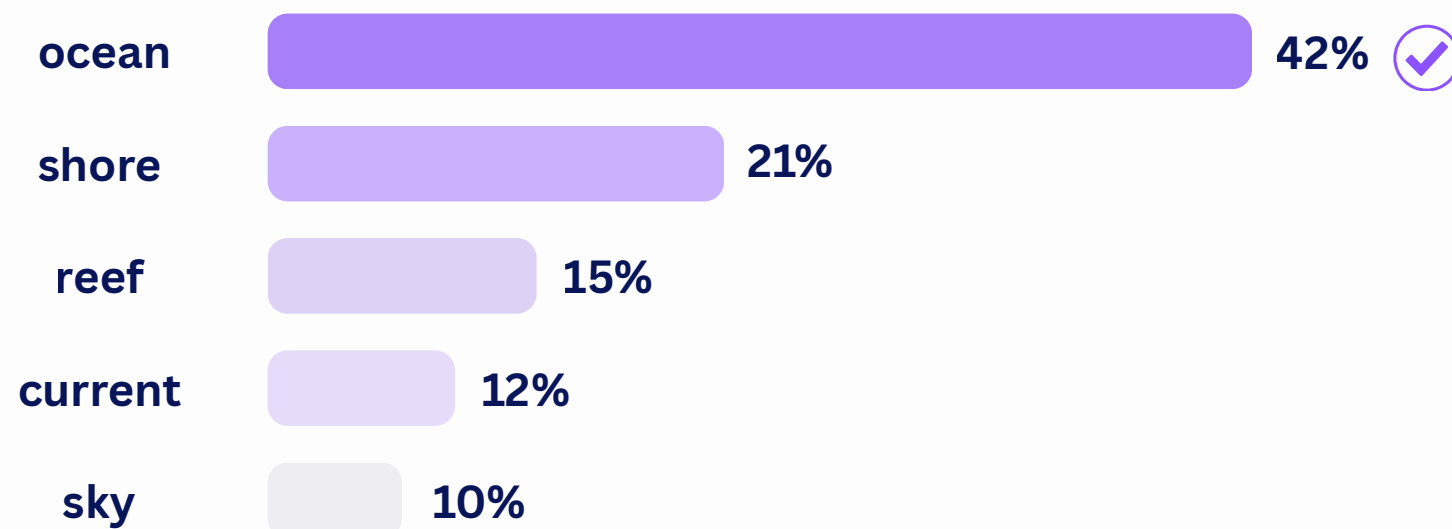
INFERENCE

A **probabilistic** system uses statistical likelihood to produce various outputs for the same input (like an LLM).

A **deterministic** system produces the same output every time for the same input (like a calculator).

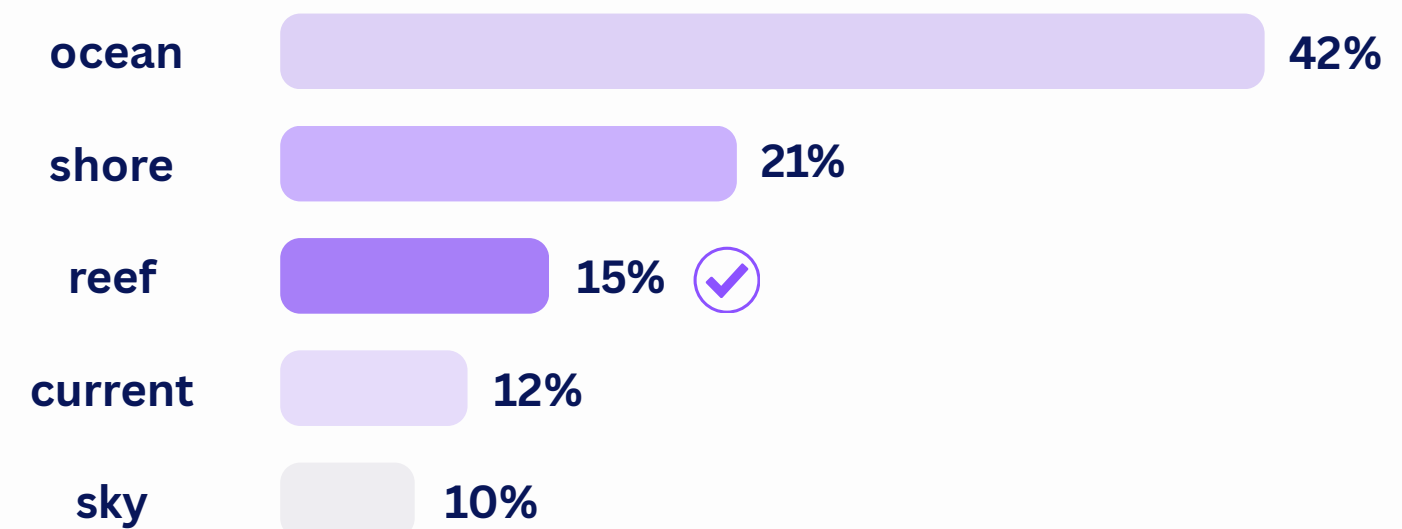
Greedy Decoding

Pick the highest probability token every time.



Sampling (Temperature 0.9)

Temperature adjusts how confident the distribution is.

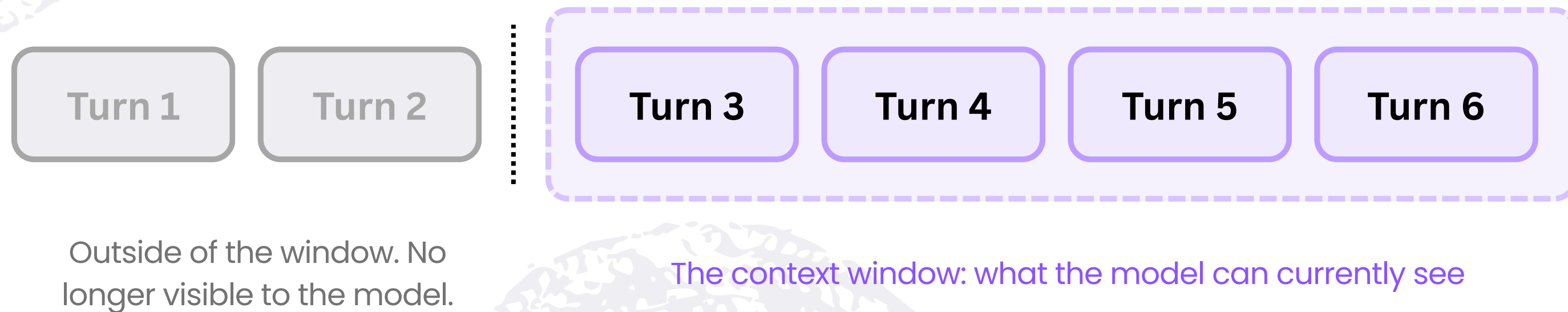


Context Windows

INFERENCE

The amount of text a model can "see" and consider at once.

This includes the prompt, your conversation so far, and any documents you've shared.



Once something falls outside the window, the model can't see it anymore.
This is why very long conversations can seem to make it "forget."

Using LLMs intelligently

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Prompts and context

What it means

When using LLMs, treat the model like a capable new hire who has vast knowledge but knows nothing about your specific situation. Be specific about audience, format, and tone, and hand it over source material instead of expecting the model to be omnipotent.

In Practice

Avoid using **weak** prompts: "Write a bio for Keiko." **Be more descriptive:** "Write a 100-word bio for Keiko, our program director, for a grant funder audience, warm but concise, drawing on the notes below."

Hallucinations

What it means

LLMs are built to predict the next plausible word, not to verify the truth. That can produce fluent, confident-sounding text that is **simply wrong**: invented citations, wrong dates, and quotes that were never said.

In Practice

Ask for the exact page number of a quote in a report, and a model may give you a specific-sounding page number and citation that does not exist. It gave an answer, not out of confidence but as a result of a probability.

Grounding and verification

What it means

Pairing a model with real, retrievable sources cuts down on made-up answers. Cross-check anything presented as fact, especially numbers, quotes, and names. **Treat AI output as a draft, not a finished fact.**

In Practice

A search-grounded assistant that shows its sources lets you click through and check the claim yourself—a plain chat answer with no receipts does not.

Bias

What it means

Training data reflects the internet's existing skews, whose voices, languages, and stories are well represented, and whose are not. A model can **reproduce and amplify those gaps**.

In Practice

A model trained mostly on English-language, Western sources may confidently answer a question about Hawaiian cultural protocol with a generic, outside guess instead of deferring to community and Native sources.

Reasoning

What it means

When a model shows "step-by-step reasoning," it is still generating **plausible-sounding text** step by step, this is not proof the model is correct, and it is not human reasoning or understanding

In Practice

A model can lay out a tidy, confident-looking chain of steps that adds up to the wrong final answer. The steps look rigorous, that alone doesn't make them right.

Human judgment and oversight

What it means

AI can draft, and it saves time, but it can also be confidently wrong. So a person always **checks the final version** before it goes out, especially anything the public sees or anything about culture or community.

In Practice

If AI drafts a social media post about a cultural practice, a human who actually knows the topic reads the post and fixes anything wrong or off before it's posted. Every time, this should be the standard – no exceptions.

Proprietary models and model families

What it means

Different companies build different models — OpenAI, Anthropic, Google — each with its own versions, or "family," released over time. Enough literacy to follow a conversation, **not to compare benchmarks.**

In Practice

"GPT-4" and "GPT-5" are versions in the same family from OpenAI — the same way "iPhone 15" and "iPhone 16" are versions from Apple. Claude and Gemini work the same way, from Anthropic and Google.